

1. Real-Time Face Recognition for Campus Security

Objective

The system identifies registered students and staff from live CCTV video while maintaining good accuracy and real-time performance.

OpenCV-Based Pipeline

Camera/Video Input → Preprocessing → Face Detection → Face Alignment → Feature Extraction → Face Recognition → Decision → Display/Record

1. Image Acquisition

- Capture live video using OpenCV VideoCapture().
- Read frames continuously from the CCTV camera.
- Resize frames to a smaller resolution to reduce computation.

2. Preprocessing

The captured image may have poor lighting or noise.

Techniques used:

- Convert BGR image to grayscale.
- Apply histogram equalization or CLAHE to improve contrast.
- Apply Gaussian filtering to reduce noise.
- Normalize the face region before recognition.

3. Face Detection

OpenCV can use:

- Haar Cascade Classifier, or
- DNN-based face detector.

The detector identifies the coordinates of faces using bounding boxes.

For example:

(x, y, width, height)

Only the detected face regions are passed to the next stage.

4. Face Alignment

Faces may appear at slightly different angles.

Alignment can be performed using facial landmarks such as:

- Eyes

- Nose
- Mouth

The face is rotated and resized to a standard size.

This improves recognition accuracy.

5. Feature Extraction

Important facial characteristics are converted into a numerical representation called a **feature vector**.

Possible approaches:

- Local Binary Patterns (LBP)
- HOG features
- Deep CNN-based face embeddings

Each registered person's features are stored in a database.

6. Recognition

The feature vector from the live camera is compared with stored feature vectors.

Possible methods:

- Euclidean distance
- Cosine similarity
- KNN classifier

If the similarity is above a predefined threshold, the person is identified.

Otherwise:

"Unknown Person"

7. Reducing False Recognition

To minimize false identification:

- Use a suitable recognition threshold.
- Require recognition in multiple consecutive frames.
- Use high-quality registered face images.
- Reject very small or low-quality faces.
- Use multiple samples for each registered person.

8. Real-Time Optimization

For low-cost computers:

- Resize video frames.
- Process every 2nd or 3rd frame.
- Detect faces only when necessary.
- Track already detected faces between detections.
- Use lightweight models.
- Use OpenCV hardware acceleration when available.

2. Intelligent Traffic Violation Detection System

Objective

The system monitors vehicles, tracks their movement, counts them, and detects violations using roadside cameras.

OpenCV Pipeline

Video Acquisition → Preprocessing → Vehicle Detection → Motion Analysis → Tracking → Line Crossing → Violation Detection → Visualization

1. Video Acquisition

- Use VideoCapture() to read CCTV video.
- Process frames continuously.
- Resize frames if necessary.

2. Preprocessing

To improve video quality:

- Convert frames to grayscale when appropriate.
- Apply Gaussian blur to remove camera noise.
- Adjust brightness/contrast.
- Use frame stabilization if the camera has small vibrations.

3. Vehicle Detection

Vehicles can be detected using:

- Background subtraction for fixed cameras.
- Contour detection for simple environments.
- OpenCV DNN/object detection models for more reliable detection.

Detected vehicles are represented by bounding boxes.

4. Motion Analysis

For a fixed camera, background subtraction can identify moving regions.

Methods include:

- MOG2
- KNN background subtraction

The foreground mask contains moving vehicles.

5. Morphological Processing

The foreground mask can contain small unwanted regions.

Use:

- **Opening** → removes small noise.
- **Closing** → fills small gaps.
- **Dilation** → strengthens detected regions.

6. Vehicle Tracking

Each detected vehicle is assigned a unique ID.

Tracking methods can maintain the identity of vehicles between frames.

For example:

Vehicle 1 → Frame 1 → Frame 2 → Frame 3

This prevents counting the same vehicle repeatedly.

7. Line-Crossing Detection

A virtual line is drawn across the road.

When the center point of a vehicle's bounding box moves from one side of the line to the other:

Vehicle Count = Vehicle Count + 1

The vehicle ID is stored so it is not counted again.

8. Detecting Abnormal Movement

The system can analyze:

- Sudden direction changes
- Wrong-way movement
- Stopping in restricted areas

- Unusual speed or trajectory
- Entering restricted regions

An alert is generated when predefined rules are violated.

9. Handling Common Problems

Shadows:

Use color/brightness analysis and morphological filtering to separate shadows from vehicles.

Camera Noise:

Gaussian or median filtering reduces noise.

Dynamic Background:

Adaptive background models such as MOG2 can adjust to moderate background changes.

Multiple Vehicles:

Use object IDs and tracking to maintain separate trajectories.

3. Automated Document Processing System

Objective

The system improves scanned documents and prepares them for accurate OCR.

OpenCV Pipeline

Document Image → Grayscale → Noise Removal → Illumination Correction → Thresholding → Morphological Processing → Segmentation → Geometric Correction → OCR

1. Image Acquisition

Scanned documents are loaded using OpenCV.

Different documents may contain:

- Noise
- Shadows
- Uneven lighting
- Blur
- Different font sizes

2. Grayscale Conversion

Convert the color document to grayscale.

Instead of processing three color channels, only one intensity channel is used.

This:

- Reduces computation.
- Simplifies thresholding.
- Makes text/background separation easier.

3. Noise Removal

Filtering removes unwanted noise.

Examples:

- Gaussian filter
- Median filter

Median filtering is particularly useful for removing small isolated noise.

4. Illumination Correction

Uneven lighting can make some parts of the document darker than others.

Techniques such as:

- CLAHE
- Background estimation
- Adaptive thresholding

can improve text visibility.

5. Thresholding

Thresholding converts the grayscale image into a binary image.

Common methods:

- Global thresholding
- Otsu thresholding
- Adaptive thresholding

Adaptive thresholding is useful when illumination varies across the document.

6. Morphological Operations

Opening

- Removes small noise.

Closing

- Connects broken parts of characters.

Dilation

- Makes text regions stronger.

Erosion

- Removes small unwanted pixels and can refine text regions.

7. Text Segmentation

The system identifies:

- Paragraphs
- Lines
- Words
- Characters

Contours and connected-component analysis can help identify text regions.

8. Geometric Correction

Scanned documents may be tilted or perspective-distorted.

The system can:

1. Detect document boundaries.
2. Find corner points.
3. Apply perspective transformation.
4. Produce a rectangular document.

This is called **deskewing/perspective correction**.

9. OCR

The cleaned and corrected image is sent to an OCR engine such as Tesseract.

Better preprocessing results in better character recognition.

10. Batch Processing

Thousands of documents can be processed automatically.

The system can:

- Read files from a folder.
- Process them one by one.
- Perform OCR.
- Save extracted text.

- Move to the next document.

4. Real-Time Multi-Person Tracking and Attendance System

Objective

The system detects students, tracks them throughout the classroom video, recognizes registered students, and records attendance only once.

OpenCV Pipeline

CCTV → Preprocessing → Face Detection → Face Alignment → Feature Extraction → Recognition → Tracking → Identity Management → Attendance

1. Video Acquisition

The classroom CCTV continuously provides video frames.

OpenCV VideoCapture() can acquire the frames.

2. Face Detection

A face detector identifies all visible faces in each frame.

For example:

Frame → Face 1 + Face 2 + Face 3 + Face 4

Each face receives a bounding box.

3. Preprocessing

For every detected face:

- Convert to grayscale if required.
- Normalize brightness.
- Resize to a standard size.
- Reduce noise.

4. Feature Extraction

Facial features are extracted from each detected face.

The system generates a feature vector for every person.

Registered students have their feature vectors stored beforehand.

5. Recognition

The current feature vector is compared with the registered database.

If similarity is sufficiently high:

Face → Student ID

Otherwise:

Face → Unknown

6. Tracking

Recognition does not need to happen on every frame.

A tracking algorithm can follow the detected face between frames.

For example:

Student A → ID 01 → Frame 1 → Frame 2 → Frame 3

This reduces computational load.

7. Handling Temporary Disappearance

A student may briefly turn away or be blocked by another student.

The tracker can maintain the student's ID for a limited number of frames.

When the face becomes visible again, the system associates it with the existing track.

8. Preventing Duplicate Attendance

Maintain an attendance list:

Student 101 → Present

Student 102 → Present

Student 103 → Present

Before marking attendance:

If Student ID already exists → Do not mark again

This guarantees that the same student is not repeatedly recorded.

9. Reducing False Recognition

Use:

- Recognition threshold.
- Multiple-frame confirmation.
- Quality checks.
- Multiple registered images.
- Tracking consistency.

A student should ideally be recognized consistently over several frames before attendance is confirmed.

10. Real-Time Optimization

- Resize video frames.
- Process selected frames.
- Track between face detections.
- Use lightweight models.
- Store only necessary feature vectors.
- Avoid repeated recognition for the same tracked person.

5. Smart Home Surveillance and Intrusion Detection

Objective

The system continuously monitors a fixed camera and generates an alert when suspicious movement occurs inside a predefined region.

OpenCV Pipeline

Camera → Preprocessing → Background Subtraction → Noise Removal → Morphology → Contour Detection → Object Tracking → Region Analysis → Alert

1. Video Acquisition

A fixed camera continuously captures video using OpenCV.

A fixed camera is useful because the background remains mostly stable.

2. Preprocessing

Each frame can be processed using:

- Resizing
- Grayscale conversion
- Gaussian blur

Blurring reduces small camera noise before motion detection.

3. Background Subtraction

The system creates a model of the normal background.

Common OpenCV methods:

- MOG2
- KNN

Moving objects appear as foreground regions.

4. Filtering

Small unwanted regions can be removed using morphological operations.

Opening: Removes small noise.

Closing: Fills gaps inside detected objects.

Dilation: Connects nearby foreground pixels.

5. Contour Detection

Contours are extracted from the foreground mask.

For each contour, calculate:

- Area
- Bounding rectangle
- Center point

Very small contours are ignored.

This prevents tiny movements from generating alerts.

6. Motion Analysis

The system examines:

- Object size
- Direction
- Speed
- Movement duration
- Position

For example, a person moving toward a door can be considered significant activity.

7. Object Tracking

A detected moving object receives a tracking ID.

Example:

Object 1 → Frame 1 → Frame 2 → Frame 3

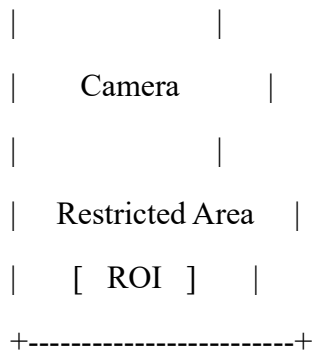
Tracking helps determine whether movement is continuous rather than random noise.

8. Predefined Region

A **Region of Interest (ROI)** is created around an entrance or restricted area.

For example:

+-----+



An alert is generated when a tracked object enters this region.

9. Reducing False Alarms

Shadows:

Use thresholding, contour properties, and brightness analysis to distinguish shadows from actual objects.

Camera Noise:

Gaussian filtering and minimum contour-area thresholds remove tiny changes.

Illumination Changes:

Adaptive background models help update the background gradually.

Trees/Moving Background:

A fixed minimum-area threshold and motion persistence checks can prevent small repetitive background movements from triggering alerts.

10. Alert Generation

When an unauthorized object enters the restricted region:

INTRUSION DETECTED

The system can:

- Display an alert.
- Record the event.
- Save the relevant frame.
- Trigger an external notification system if connected.

11. Real-Time Optimization

For low-cost computers:

- Reduce frame resolution.
- Process selected frames.
- Use efficient background subtraction.

- Track objects instead of detecting them repeatedly.
- Process only the required ROI.